



ROOT CAUSE UNLOCKED · RECOVERY GUIDE

The Neck and Shoulder *Recovery* Guide

The in-clinic tissue prep protocols, the movement evidence, and the honest supplement picture for neck pain, cervicogenic headache, rotator cuff pain, and tennis elbow

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The Neck and Shoulder Recovery Guide

BEFORE YOU START

Contents

Read this part first

Why one guide covers the neck, shoulder, and elbow

If you were told you have... (your diagnosis, translated)

The one idea that makes this kit honest

The neck sequence

The shoulder sequence

The elbow and forearm sequence (the tennis elbow work)

The program

When to get evaluated instead of self-treating

Why nutrient evidence looks thinner than it is

Sort the tissue before you buy anything

If your problem is tendon: the biology, and what follows from it

If your problem is neck pain or headache: the shelf really is close to empty

The floor under all of it

What I would actually run, and how to judge it

Interactions, and the ones people miss

Track it, so the program can be judged by data

Dr. Seay recommends: the exact products

Working with Root Health

References

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Some neck, shoulder, and arm pain is an emergency. Know these before anything else in this guide:

- Shoulder, arm, or jaw discomfort that comes on with exertion, or arrives with chest pressure, sweating, nausea, or breathlessness. Left shoulder and arm pain can be a heart attack presenting politely. **Call 911.** This is the single most important sentence in this guide.
- A sudden, severe headache unlike any you have had, or neck stiffness with fever. Emergency evaluation.
- Neck pain after real trauma (a crash, a fall). Do not massage it; get imaged.
- Progressive arm weakness or numbness, new clumsiness in the hands, or a change in your walking. These can signal spinal cord involvement and need prompt medical evaluation, not a massage gun.
- Neck pain arriving with dizziness, double vision, slurred speech, or trouble swallowing: urgent evaluation.

The tool safety rules, from the same sheets we use in clinic: nothing over an acute injury in its first 48 to 72 hours; reduce pressure over inflamed areas; skip percussion devices entirely with a pacemaker or other implanted electronic device; avoid aggressive pressure over varicose veins or compromised skin; caution on blood thinners. This region adds nerve-specific rules from the clinic sheets: no direct pressure on the bony elbow points (epicondyles), stay aware of the ulnar nerve at the inner elbow, reposition immediately off any spot that produces numbness or tingling, and stop entirely if neurological symptoms appear. Start light. On the manual tools, the foam roller, roller stick, ball and trigger point tool, where the pressure is your own body weight, pain must not exceed 6 out of 10: uncomfortable but tolerable. **On a powered percussion device the rule is different and stricter.** Follow the manufacturer's instructions, which say to use it without producing pain or discomfort and to stop immediately if you feel any, apart from light muscle soreness. I am not asking you to push a powered tool to a number. Sharp or shooting pain means back off, whatever the tool.

Nothing here is a reason to stop a prescription or skip a recommended evaluation.

Why one guide covers the neck, shoulder, and elbow

These three share a platform: the shoulder blade. Deep neck flexor endurance, scapular positioning, and mid-back mobility drive all three regions, a headache that starts in the neck, a rotator cuff that gets pinched, and an elbow tendon that overloads because the shoulder upstream is not doing its share. They also share tools, which is why your clinic sheets group the same devices across all three. One guide, one region, one program.

The mid-back is the part most people never think to blame, and it is the reason this guide exists. Your mid-back is the thoracic spine, the twelve vertebrae your ribs attach to, and it is built to be your most mobile stretch of spine. Your neck and your shoulder blade are built to be stable. When a joint that is supposed to move stops moving, the joints above and below it end up moving more than they were designed to, and those are the ones that start hurting. So a stiff thoracic spine does not usually announce itself as mid-back pain. It shows up as a neck that aches by afternoon, or a shoulder that catches when you reach overhead.

This is why so much of what follows works on the area between your shoulder blades even when your complaint is somewhere else. It is also why the foam roller work below and “thread the needle” are in your sheets: one restores thoracic extension, the other restores thoracic rotation.

Two honest notes on the evidence. Treating the thoracic spine for neck pain has been tested, and a meta-analysis of randomized trials found immediate improvements in pain, cervical mobility, and function (PMID 41987777). A broader systematic review of thoracic treatment across upper quadrant problems, meaning neck and shoulder together, is more mixed and graded the evidence carefully (PMID 41196244). So this is a well-reasoned principle with real support for the neck, and thinner support for the shoulder. I use it either way, because restoring motion where motion belongs costs you nothing and the alternative is chasing the site of the pain forever.

If you were told you have... (your diagnosis, translated)

- **“Lateral epicondylitis” or “lateral epicondylalgia”:** tennis elbow. Same condition, and the elbow section is yours.
- **“Golfer’s elbow” (medial epicondylitis):** the inner-elbow cousin. The elbow protocol in part one already treats both sides of the elbow, and the load-it-progressively principle transfers, though the trial evidence was built mostly on the lateral side.
- **“Shoulder impingement,” “subacromial pain,” or “shoulder bursitis”:** the rotator cuff territory, and the CSAW and GRASP trials in part two are specifically about it.
- **A “rotator cuff tear” on MRI:** imaging finds cuff changes in plenty of pain-free shoulders past middle age, so a tear on a report is not automatically the pain source. Bring the report and this guide’s part two to your clinician; the answer is individual.
- **“Frozen shoulder” (adhesive capsulitis):** a genuinely different condition with its own timeline and plan. A shoulder progressively losing motion in every direction needs diagnosis, not this program; the companion ebook *The Complete Patient’s Guide to Frozen Shoulder* covers that territory, including the surgery trial and the diabetes connection.
- **“Text neck,” “military neck,” or “straightening of the cervical curve”:** imaging descriptions, not diagnoses. The strengthening approach in part two is the same.

The one idea that makes this kit honest

The tools buy you a window, and the movement spends it. Foam rolling used as a warm-up produces small improvements in flexibility (PMID 31024339); massage guns improve short-term range of motion and recovery measures but nothing lasting, and are not recommended immediately before demanding strength or balance work (PMID 37754971). That last point sets the order of your session rather than its contents, so to be specific rather than leave you to reconcile it: the foam roller, the trigger point tool, the heat wrap and the mobility drills can run straight into your strengthening, and the percussion massager should not. Use it after a session, on a rest day, or leave roughly half an hour before you load. The tools make the region feel workable so the strengthening can happen. The strengthening is the treatment, and in this region that is not a slogan; it is the repeated finding of the trials in part two, where loading programs matched or beat both surgery and injections.

Part one: the tissue prep protocols, exactly as used in clinic

Do these before your movement work, not instead of it. Respect the pressure ceilings above, manual versus powered, and the nerve rules throughout.

One timing rule carries over from the section above, and it is the only place the order matters: the percussion massager and the vibrating ball are the tools not to use immediately before strength work. Everything else here can run straight into your session.

The neck sequence

- **Vibrating massage ball**, vibration on, along the upper trap: roll back and forth over tender and tight areas. 2 to 3 minutes. **Or a massage gun**, attachment of your choice, over the same area for 2 to 3 minutes.
- **Neck trigger point tool**: work up and down the neck for 2 to 3 minutes, focusing on tight and tender areas, moving the handles side to side and up and down to work the tissue.
- **Foam roller** between the shoulder blades: lift your buttocks off the floor, one hand behind your head, and roll up and down. 1 to 2 minutes.

The shoulder sequence

- **Foam roller** between the shoulder blades, rolling a little past the ends of the blades. 1 to 2 minutes.
- **Heat and vibration shoulder wrap, set to vibrate and heat, worn while performing:**
- **90/90 pillow press:** arm at 90 degrees, drive the hand down toward the ground, hold 3 to 5 seconds, return; then rotate back toward the head, back of the hand reaching for the ground, hold 3 to 5 seconds. Keep the arm at 90 degrees. 5 reps each direction, each shoulder.
- **Reach, roll, and lift:** kneeling with arms on an exercise ball, palms up, roll the ball forward until you feel the stretch in the lats. Hold 5 to 8 seconds, 5 repetitions.
- **Thread the needle:** on hands and knees, rotate one arm up and back as far as comfortable, then sweep it across the body under the other arm, legs staying put. Hold 3 to 5 seconds, 5 reps each side.
- **Massage gun** (round or flat head) in the armpit, elbow bent: move the arm overhead and back down. 1 minute each side. If you feel numbness or tingling, reposition off the nerves.
- **Massage ball** on the back of the shoulder for the infraspinatus and teres minor. 1 minute on each muscle.
- **Massage ball** pinned between your pec muscle and a wall: move the arm up and down and away from the wall while working tender areas.

The elbow and forearm sequence (the tennis elbow work)

- **Massage gun:** triceps (round ball or dampener head, level 2 to 3, circular motions, 60 to 90 seconds, avoiding the elbow joint itself); biceps and brachialis (round ball, level 2, gentle, 45 to 60 seconds, avoiding the bicipital groove); forearm flexor mass (dampener or fork, level 1 to 2, linear strokes along the inside forearm, 45 to 60 seconds, light pressure near the inner bony point); forearm extensor mass (dampener, level 1 to 2, linear strokes along the outside forearm, 45 to 60 seconds); brachioradialis (round ball, level 1 to 2, 30 to 45 seconds along the muscle belly).
- **Massage ball,** pressed with the opposite hand: gentle circles around the outer elbow point for 60 to 90 seconds, focusing on the extensor tendon origin, off the bone itself, watching for nerve irritation; the same gently around the inner elbow point; then up and down the outer and inner intermuscular septum lines of the upper arm, 60 to 90 seconds each.
- **Firm trigger point tool,** linear strokes: triceps shoulder-to-elbow (45 to 60 seconds); biceps and front of the upper arm (45 to 60 seconds); forearm flexors elbow-to-wrist on the palm side (30 to 45 seconds); forearm extensors on the back of the forearm (30 to 45 seconds); brachioradialis along its line, following it above the elbow (30 to 45 seconds).

Part two: the movement evidence, which is the actual treatment

This region's trials share one repeated, uncomfortable finding: the patient interventions beat the procedural ones.

For neck pain, strengthen; do not just stretch. The Cochrane review of exercise for mechanical neck disorders found no high-quality evidence anywhere in this literature, and said so, but its consistent signal is that specific strengthening and endurance work for the neck, shoulder blade region, and shoulder may reduce pain and improve function in chronic neck pain, cervicogenic headache, and radiculopathy, while stretching alone showed no expected benefit (PMID 25629215). If your neck program is only stretches, it is the half of the program the evidence does not support.

For headaches that start in the neck, the trial worth knowing tested both of the things this practice does. A multicenter randomized trial of cervicogenic headache compared manipulative therapy, a low-load strengthening exercise program, and their combination against control: at 12 months, both manipulative therapy and specific exercise had significantly reduced headache frequency and intensity, effects were maintained, and the combination gave relief to about 10 percent more patients (PMID 12221344). Chiropractic care and the home program in this guide are teammates in that trial, not rivals.

For rotator cuff related shoulder pain, two large trials reset the conversation. In the CSAW trial, arthroscopic subacromial decompression, one of the most common shoulder surgeries, produced outcomes no better than placebo arthroscopy, and the difference versus no treatment was not clinically important; the authors state plainly that the findings question the value of the operation for this indication (PMID 29169668). In the GRASP trial of over 700 patients with rotator cuff disorders, a progressive exercise program was not superior to a single best-practice physiotherapy advice session with a home program, and corticosteroid injection provided no long-term benefit (PMID 34265255). Read those together honestly: the active ingredient is graded loading of the shoulder, it does not require an operating room, and even a modest, consistent home program is in the evidence's winning column.

For tennis elbow, the injection finding needs to be read twice, because it runs backwards. In a randomized trial published in JAMA, corticosteroid injection produced *worse* outcomes at one year than placebo injection (PMID 23385272). An earlier BMJ trial found the same paradox: injections win the first six weeks, then reverse, with high recurrence rates, while physiotherapy (elbow mobilisation plus exercise) beat wait-and-see early and beat injections from six weeks on (PMID 17012266). Most tennis elbow also improves within a year regardless, which is worth knowing before paying for anything. The honest sequence: load it progressively, be patient, and treat the quick-fix needle as the option with the best week-two results and the worst year-one results in its own trials.

The program

Start with the most useful thing the trials in this region found, because it is not what you would expect.

In GRASP, over 700 people with rotator cuff pain were given either a progressive, supervised exercise programme or a single best-practice advice session with a home programme, and **the elaborate version was not superior** (PMID 34265255). In this region, a modest home program you actually do is in the winning column. That is the opposite of what a supervised package is usually sold on, and it is the reason this program is short enough to finish.

The other repeated theme: strengthen, do not only stretch. The neck review found that specific strengthening and endurance work may reduce pain and improve function, while stretching alone showed no expected benefit (PMID 25629215). If your current program is only stretches, you are doing the half the evidence does not support.

The core, for everybody in this guide

1. **Strength, two or three times a week, on non-consecutive days.** Two to three sets of 8 to 12 repetitions, slow and controlled. Slow is the technique.
2. **Thoracic mobility most days**, meaning the foam roller work and thread the needle from part one. This is the one piece of mobility work that earns its place here, for the reasons in the thoracic section above.
3. **Patience measured in months.** Every trial in this part measured its wins at 6 and 12 months, not at two weeks.

Start here: pick the one that sounds like you

If your picture is neck pain:

- Deep neck flexor work is the priority: lying on your back, gently nod as though saying yes, drawing the chin down and back without lifting the head, and hold. Build holding time before you build anything else. This is endurance work, not a strength contest, and it should never be forceful.
- Add scapular work: rows, and shoulder blade squeezes held rather than snapped.
- Add the thoracic mobility from part one.
- Stretching is a comfort measure here, not the treatment.

If your picture is headaches starting in the neck:

- The same low-load neck endurance work above is your program, and it is the one that was tested. In the trial, both manipulative therapy and a low-load exercise program reduced headache frequency and intensity, the effects were still there at 12 months, and combining them helped about ten percent more people (PMID 12221344).

- That trial is the reason care in my office and your home program are teammates rather than alternatives. If you are doing one, the evidence says add the other.
- Low load is the whole point. Headache programs fail when people turn a gentle endurance drill into a hard one.

If your picture is shoulder pain from the rotator cuff:

- Graded loading is the active ingredient, and it does not need an operating room. Arthroscopic decompression performed no better than placebo surgery for this problem (PMID 29169668).
- Start with the range and load you can control without a pain spike, and add gradually. Rows, external rotation with a band, and controlled work below shoulder height first.
- Because the simple version was not beaten by the elaborate one, err toward something you will do three times a week for three months.
- **A shoulder losing motion in every direction is a different condition** and needs diagnosis rather than this program.

If your picture is tennis elbow:

- Slow, heavy wrist extensor loading is your program. Lower the weight slowly, which is the part that matters most, and build load over weeks.
- Be patient on purpose. Most tennis elbow improves within a year regardless, and the injection that wins at week two produced worse outcomes at one year than a placebo injection (PMID 23385272, PMID 17012266). Knowing that before you pay for anything is the point.
- Expect this to be the slowest pathway in the guide. That is the condition, not your effort.

How to make it harder

Progress one thing at a time, roughly every one to two weeks, and only when the current version is comfortable: more repetitions first, up to about 12, or longer holds for the neck endurance work; then a harder version; then more load; then more sets. If you add difficulty and the next two sessions feel worse, drop back a step for a week.

How to know it is going right, and when to back off

The soreness rule. Discomfort during and after is expected and is not damage. The test is the next morning. Settled by then means carry on. Still there or worse means the last session was too much, so return to the version you handled comfortably.

Stop immediately and reposition if any tool or exercise produces numbness, tingling, or shooting pain down the arm. That is a nerve, not a muscle, and it is never something to push through.

Stop and get examined for anything in the warning list at the top of this guide, for night pain that will not settle, for a shoulder losing motion in every direction, for any progressive weakness or numbness, or for headaches that change character.

Give it twelve weeks before you judge it, and check in at six. Six weeks of honest effort with nothing to show is a reason to be examined rather than a reason to push harder.

When to get evaluated instead of self-treating

Beyond the emergencies on page one: pain not trending better by six weeks of honest effort, night pain that will not settle, a shoulder losing motion in every direction (frozen shoulder pattern, which needs its own plan), any progressive weakness or numbness, or headaches changing character. An examination beats a guide.

Part three: the supplement half, honestly graded

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food, so the evidence hierarchy overdocuments patentable drugs and under-documents nearly everything else. A Cochrane methodology review found manufacturer sponsorship produces more favourable results and conclusions than independent funding, through a bias standard quality checks do not catch (PMID 28207928). So when this guide says “no evidence,” it will say which kind: *tested and failed*, *never properly tested*, or *tested small and won*. The asymmetry earns nutrients humility, not automatic credit; small nutrient trials carry their own publication bias toward positive findings. One rule, both directions, and I apply it to the supplement industry below exactly as hard as I apply it to drug companies.

Sort the tissue before you buy anything

It would be easy to tell you this shelf is nearly empty and leave it there. That is half right, and it is not useful, so here is the better version.

Most of what hurts in this region is **tendon**. Rotator cuff pain is tendon. Tennis elbow is tendon. That matters enormously, because tendon is not a joint and it does not want what the joint-supplement aisle is selling. It is a rope of collagen, it remodels slowly, and it is rebuilt by loading it. There is real biology here and a real protocol that follows from it, and no joint capsule is any part of it.

Neck pain and headache are mostly muscle, joint and nerve rather than tendon, and there the shelf genuinely is close to empty. I will say so plainly rather than sell you the tendon protocol twice.

One rule holds either way, and it is why this section sits behind the movement section. Every trial worth citing below layered its supplement on top of a loading program. Nutrition supplies raw material. Load is what tells the body where to put it. A supplement taken instead of the program is money spent on a signal your body has no reason to act on.

If your problem is tendon: the biology, and what follows from it

Start with what a tendon actually is. Roughly three quarters of the dry weight of tendon is type I collagen, a rope of protein wound from repeating glycine, proline and lysine. To build it, your body chemically modifies proline and lysine after the chain is assembled, and the enzymes that do that work, prolyl hydroxylase and lysyl hydroxylase, cannot function without vitamin C and iron. The enzyme that then cross-links those ropes into something that carries load, lysyl oxidase, requires copper. Manganese and zinc are required elsewhere on the same assembly line. This is textbook connective tissue biochemistry rather than a supplement claim, and it is the reason a collagen program that supplies protein without supplying cofactors is only doing half the job.

Now the timing, which is the part almost nobody gets right. Collagen synthesis in tendon is driven by mechanical load and rises in the hours after you load the tissue. So the useful question is not “should I take collagen,” it is “is the raw material in my bloodstream at the moment my tendon is being told to rebuild.” In the study that established this, eight healthy men took vitamin C-enriched gelatin one hour before six minutes of rope skipping. The 15 gram dose doubled a blood marker of new collagen formation, and serum drawn from those subjects improved the collagen content and mechanics of engineered ligaments in the lab (PMID 27852613).

A later trial changed the dose I would give you, and it is worth more than any label on a tub. A dose-response study found that 30 grams of hydrolyzed collagen with 50 mg of vitamin C, taken an hour before heavy resistance exercise, produced more whole-body collagen synthesis than 15 grams, while **15 grams was not statistically different from nothing at all** (PMID 38007183). Most collagen products on the shelf, and most people taking them, sit at or below the dose that failed to beat placebo. If you are going to run this experiment, run it at a dose the research actually supports, and take it about an hour before the session rather than at bedtime.

Here is the ceiling on that claim, and I would rather you hear it from me. A blood marker of collagen synthesis is not a healed tendon. When an independent group tested whether collagen peptides change tendon structure, giving 15 grams daily through fifteen weeks of resistance training and imaging the patellar tendon with MRI, they found no advantage over placebo on tendon size, stiffness or mechanical properties. Both groups improved, and the training did the work (PMID 37436929). The most recent independent systematic review and meta-analysis, nineteen studies and 768 participants, found small statistically significant effects favoring collagen peptides and graded the certainty of the tendon findings specifically as very low (PMID 39060741). There is also a genuinely positive symptom trial: 139 athletic adults with activity-related joint pain took 5 grams of bioactive collagen peptides daily for twelve weeks and reported significantly less activity-related pain than placebo, with less need for other treatments (PMID 28177710). That trial carries a correction notice and one of its authors works for a collagen research institute, which is exactly the

sponsorship pattern the Cochrane review at the top of this section warns about. I cite it anyway, labeled, because pretending it does not exist would be its own kind of dishonesty.

So the fair summary is this. The mechanism is real and well described. The acute response is real and dose-dependent. Symptom trials are positive but small and mostly industry-adjacent. The one hard structural endpoint tested independently came back null. That is a reasonable, cheap, low-risk experiment with honest odds, and it is not a treatment with proven structural benefit. Anyone selling it to you as the latter is ahead of the evidence.

The cofactor question is the smarter half of it. If the substrate is only useful when the assembly line is staffed, then the cofactors matter as much as the grams of collagen, and they are the part almost every consumer product ignores. A practitioner formula supplying vitamin C, copper, manganese, zinc and the amino acids proline and lysine together is built on real biochemistry rather than a marketing story. Two honest notes before you buy one. Its amino acid content is modest, typically around 150 mg each of proline and lysine, which is a rounding error next to the 15 to 30 grams used in the collagen trials, so it complements a collagen powder rather than replacing it. And these formulas often contain iron, which is a genuine cofactor here, but most men and most postmenopausal women should not take supplemental iron without a documented reason, because iron is one of the few nutrients where routinely taking more than you need is actively harmful.

Stage matters more than product, and this is the most useful paragraph in the section. In a three-arm randomized trial of 59 patients with Achilles tendinopathy, a supplement of mucopolysaccharides with type I collagen and vitamin C added benefit specifically in patients with early, reactive tendinopathy, and added nothing once the tendon had reached the degenerative stage with established matrix and vascular change (PMID 28053674). That trial was run in an Achilles rather than a cuff or an elbow, so treat the specific result as borrowed. What transfers is the principle, and it is a good one: an early, angry, reactive tendon is still responsive tissue, and that is the window where nutritional support appears to add something, while a long-standing degenerative tendon has structural change a capsule did not reverse. This is why “how long has this been going on” is one of the first questions I ask, and it should change what you spend money on. If your shoulder or elbow pain is weeks old, the supplement side is worth running. If it is two years old, put the money and the attention into the progressive loading program in part two, because that is what has actually been shown to change these tissues at that stage.

If your problem is neck pain or headache: the shelf really is close to empty

There is no supplement with convincing outcome trials for neck pain or for cervicogenic headache, and that is mostly *never properly tested* rather than *tested and failed*. I am not going to fill that gap with borrowed tendon evidence, because the neck's problem is not a collagen substrate problem.

What I will not do is end the paragraph there. For this group the leverage is entirely in the things part two describes, deep neck flexor endurance and scapular work, plus the two unglamorous items at the end of this section that apply to everyone. If you came to this section hoping for a capsule for your neck, the honest answer is that the money is better spent on the loading program and on sleep, and I would rather tell you that than take the sale.

The floor under all of it

The interventions with the best evidence in this entire section are not sold in the supplement aisle.

Protein is the one number worth knowing. The meta-analysis and meta-regression across 49 studies and 1,863 participants found that protein supplementation meaningfully improves gains from resistance training, and that benefits plateau around 1.6 grams per kilogram of body weight per day (PMID 28698222). For a 180 pound person that is roughly 130 grams daily. Most people rehabbing an injury eat well under that, and it is the cheapest correction available to you. Collagen specifically is a poor quality protein in isolation, low in the essential amino acids muscle needs, so treat a collagen dose as a targeted addition to your protein intake and never as your protein intake itself.

One disclosure on that paper, because I said I would run this rule in both directions. It carries a later correction noting that one of its authors sat on a sports supplement manufacturer's advisory board while it was written. The correction changed none of the data and the 1.6 figure stands, but holding the supplement industry to the standard I hold drug companies to means saying so when the study behind a number I like has a commercial tie on it, and not only when the number is one I dislike.

Sleep is when tissue repair actually happens, and energy sufficiency is what funds it. Smoking, meanwhile, is genuinely bad for tendon healing, and quitting outranks any capsule this aisle sells. Those items outrank everything in a bottle and they earn me nothing.

What I would actually run, and how to judge it

Pick the tissue, not the product. If your problem is a cuff or an elbow tendon, the defensible experiment is 15 to 30 grams of gelatin or hydrolyzed collagen with vitamin C taken about an hour before your loading sessions, with cofactors covered, run for twelve weeks, and weighted toward the early-and-reactive case rather than the long-standing one. If your problem is neck pain or headache, buy nothing here and put the effort into part two. Underneath either, hit 1.6 grams of protein per kilogram, sleep, and eat enough.

Then judge it. Set a start date and an end date before you buy anything, change one thing at a time, and write down what you are measuring. Everything in this section is a defined trial with a stop date, not a subscription. If twelve weeks of honest effort produced nothing, that is a real answer and it is worth knowing, because it sends you back to the loading program and to an examination instead of to the next bottle.

Interactions, and the ones people miss

Do not add supplemental iron, including inside a combination formula, without a reason to. Proprietary blends hide the dose of each ingredient behind a single total, which makes it impossible to know whether you are getting a researched amount of anything, so prefer products that disclose amounts. And whichever source you use, look for independent third-party testing, because the manufacturing side of this industry is even less regulated than the marketing side.

Nothing in this section substitutes for being examined, and none of it applies to pain that is getting worse, waking you at night, or accompanied by any of the warning signs at the top of this guide, the cardiac referral especially.

Track it, so the program can be judged by data

The free Root Cause Unlocked tracker has an MSK mode built for this guide: daily function, range of motion, load tolerated, and flare days. Eight weeks of that turns “I think it might be helping” into an answer.

Free tracker: [rootcauseunlocked.com/tracker?](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=msk-neckshoulder-recovery)

[utm_source=guide&utm_medium=ebook&utm_campaign=msk-neckshoulder-recovery](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=msk-neckshoulder-recovery)

Dr. Seay recommends: the exact products

The protocols in this guide are written generically on purpose, because the technique matters and the brand mostly does not. A lacrosse ball works. This section is the other half of that promise: if you would rather not stand in an aisle guessing, here is exactly what I would hand you, why that specific item and not a cheaper-looking equivalent, and where to get it. Read the disclosure at the end before you buy anything, because I want you deciding with your eyes open.

On the supplement side, through my dispensary

This is a short list on purpose. Most of this region's leverage is in part two, and I am not going to pad the list to make the section look substantial.

us.fullscript.com/welcome/drseay

For collagen substrate, if a cuff or elbow tendon is your picture: a plain hydrolyzed collagen or gelatin powder, 15 to 30 grams, with vitamin C. Here the brand genuinely does not matter and I will not pretend otherwise. What matters is the dose and the timing: aim toward the upper end of that range, take it about an hour before your loading session, and make sure vitamin C is on board. An unflavored bulk powder is the honest buy. Do not pay a premium for a proprietary blend delivering two grams, which is the most common way people spend money on this idea and get nothing back.

For collagen cofactors: Collagenics, by Metagenics. Three tablets daily. This is the assembly line rather than the raw material: vitamin C, copper, manganese, zinc and the amino acids proline and lysine in one formula, which is the cofactor set the biochemistry section above describes. Read that section's iron caution first. Collagenics contains 6 mg of iron, and if you are a man or a postmenopausal woman without documented low iron, ask me or your prescriber before starting it rather than after.

For an early, reactive tendon: SynovX Tendon and Ligament, by Xymogen. Two capsules daily. This supplies the mucopolysaccharide and type I collagen combination with vitamin C tested in the tendinopathy trial above, where it added benefit in early-stage reactive tendinopathy and not in long-standing degenerative tendons (PMID 28053674). That trial was run in an Achilles rather than a cuff or an elbow, so match it to your stage rather than assuming it was proven in your shoulder. If your tendon pain is years old, spend the money on the loading program instead, and I would rather tell you that than sell you a bottle.

If neck pain or headache is your picture, buy none of the above. The evidence does not support it and I am not going to sell you something so the section feels complete. Put the money toward protein and toward the loading program in part two.

On the tool side

The protocols name categories, not brands, and every one can be done with inexpensive equipment. Any dense foam roller does the rolling work, a vibrating roller just adds vibration. A massage ball or lacrosse ball substitutes for a vibrating sphere. The percussion massager, the heat-plus-vibration shoulder wrap and the neck trigger point tool are the three items without a close household substitute, and they are also the most expensive, so they are the ones worth thinking hardest about.

If you want the specific equipment I use in the clinic, these are the links:

Percussion massager, vibrating roller, vibrating sphere and heat-plus-vibration shoulder wrap:
[[HYPERICE_AFFILIATE_LINK]] Foam roller, massage ball, neck trigger point tool, firm trigger point tool and exercise ball: [[GENERAL_TOOLS_AFFILIATE_LINK]]

Disclosure: Root Health earns a portion of dispensary purchases, and the tool links above are affiliate links, which means we may earn a commission if you buy through them at no additional cost to you. Nothing in this guide was included, excluded, ranked, or dosed based on what it earns. The generic substitutions listed above are real substitutions and I would rather you use a lacrosse ball than skip the work, and the highest-value items in this guide, the loading program, protein, and sleep, earn us nothing at all.

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your history, your exam, and your actual loading tolerance.

***Book a free Root Discovery Session: [rootcauseunlocked.com/discovery?](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=msk-neckshoulder-recovery)
[utm_source=guide&utm_medium=ebook&utm_campaign=msk-neckshoulder-recovery](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=msk-neckshoulder-recovery)***

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

***Get in touch: [rootcauseunlocked.com/contact?](https://rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=msk-neckshoulder-recovery)
[utm_source=guide&utm_medium=ebook&utm_campaign=msk-neckshoulder-recovery](https://rootcauseunlocked.com/contact?utm_source=guide&utm_medium=ebook&utm_campaign=msk-neckshoulder-recovery)***

The companion ebooks go deeper on the conditions in this region: *The Complete Patient's Guide to Rotator Cuff Shoulder Pain* and *The Complete Patient's Guide to Tennis Elbow*.

References

References 1 to 12 were verified against live PubMed records on 2026-08-09. References 13 to 20, added with the rebuilt supplement section, were verified against live PubMed records on 2026-09-04. Both passes included a publication type and corrections check for retractions and errata. On 2026-09-05 every reference was additionally screened for superseded editions; none in this guide is superseded. That same screen surfaced a previously undisclosed correction on reference 19, noted there. Verification detail is documented in `Recovery-Guide-Citation-Ledger.md`.

1. Gross A, Kay TM, Paquin JP, et al. Exercises for mechanical neck disorders. *Cochrane Database Syst Rev*. 2015;1(1):CD004250. PMID 25629215
2. Jull G, Trott P, Potter H, et al. A randomized controlled trial of exercise and manipulative therapy for cervicogenic headache. *Spine*. 2002;27(17):1835-1843. PMID 12221344
3. Beard DJ, Rees JL, Cook JA, et al. Arthroscopic subacromial decompression for subacromial shoulder pain (CSAW): a multicentre, pragmatic, parallel group, placebo-controlled, three-group, randomised surgical trial. *Lancet*. 2018;391(10118):329-338. PMID 29169668
4. Hopewell S, Keene DJ, Marian IR, et al. Progressive exercise compared with best practice advice, with or without corticosteroid injection, for the treatment of patients with rotator cuff disorders (GRASP): a multicentre, pragmatic, 2 x 2 factorial, randomised controlled trial. *Lancet*. 2021;398(10298):416-428. PMID 34265255
5. Coombes BK, Bisset L, Brooks P, Khan A, Vicenzino B. Effect of corticosteroid injection, physiotherapy, or both on clinical outcomes in patients with unilateral lateral epicondylalgia: a randomized controlled trial. *JAMA*. 2013;309(5):461-469. PMID 23385272
6. Bisset L, Beller E, Jull G, et al. Mobilisation with movement and exercise, corticosteroid injection, or wait and see for tennis elbow: randomised trial. *BMJ*. 2006;333(7575):939. PMID 17012266
7. Shaw G, Lee-Barthel A, Ross ML, Wang B, Baar K. Vitamin C-enriched gelatin supplementation before intermittent activity augments collagen synthesis. *Am J Clin Nutr*. 2017;105(1):136-143. PMID 27852613
8. Wiewelhove T, Döweling A, Schneider C, et al. A Meta-Analysis of the Effects of Foam Rolling on Performance and Recovery. *Front Physiol*. 2019;10:376. PMID 31024339
9. Ferreira RM, Silva R, Vigário P, et al. The Effects of Massage Guns on Performance and Recovery: A Systematic Review. *J Funct Morphol Kinesiol*. 2023;8(3). PMID 37754971
10. Lundh A, Lexchin J, Mintzes B, et al. Industry sponsorship and research outcome. *Cochrane Database Syst Rev*. 2017;2(2):MR000033. PMID 28207928
11. Pan Y, Chen J, Liu Y, Chen Z, Liu J. The immediate effects of thoracic spine manipulation in patients with neck pain: a meta-analysis of randomized controlled trials. *Front Med (Lausanne)*. 2026;13:1790614. PMID 41987777
12. Thoomes EJ, Tilborghs G, Heneghan NR, Falla D, De Graaf M. Effectiveness of thoracic spine manipulation for upper quadrant musculoskeletal disorders: a systematic review. *J Manipulative Physiol Ther*. 2025;48(1-5):422-434. PMID 41196244
13. Shaw G, Lee-Barthel A, Ross ML, Wang B, Baar K. Vitamin C-enriched gelatin supplementation before intermittent activity augments collagen synthesis. *Am J Clin Nutr*. 2017;105(1):136-143. PMID 27852613
14. Lee J, Tang JCY, Dutton J, et al. The Collagen Synthesis Response to an Acute Bout of Resistance Exercise Is Greater when Ingesting 30 g Hydrolyzed Collagen Compared with 15 g and 0 g in Resistance-Trained Young Men. *J Nutr*. 2024;154(7):2076-2086. PMID 38007183

15. Balshaw TG, Funnell MP, McDermott EJ, et al. The Effect of Specific Bioactive Collagen Peptides on Tendon Remodeling during 15 wk of Lower Body Resistance Training. *Med Sci Sports Exerc.* 2023;55(11):2083-2095. PMID 37436929
16. Bischof K, Moitzi AM, Stafilidis S, Konig D. Impact of Collagen Peptide Supplementation in Combination with Long-Term Physical Training on Strength, Musculotendinous Remodeling, Functional Recovery, and Body Composition in Healthy Adults: A Systematic Review with Meta-analysis. *Sports Med.* 2024;54(11):2865-2888. PMID 39060741
17. Zdzieblik D, Oesser S, Gollhofer A, Konig D. Improvement of activity-related knee joint discomfort following supplementation of specific collagen peptides. *Appl Physiol Nutr Metab.* 2017;42(6):588-595. PMID 28177710 (Erratum: *Appl Physiol Nutr Metab.* 2017;42(11):1237. Corrected, not retracted. One author is affiliated with a collagen research institute, disclosed in the text.)
18. Balias R, Alvarez G, Baro F, et al. A 3-Arm Randomized Trial for Achilles Tendinopathy: Eccentric Training, Eccentric Training Plus a Dietary Supplement Containing Mucopolysaccharides, or Passive Stretching Plus a Dietary Supplement Containing Mucopolysaccharides. *Curr Ther Res Clin Exp.* 2016;78:1-7. PMID 28053674 (Two authors are employed by the ingredient manufacturer, disclosed in the text.)
19. Morton RW, Murphy KT, McKellar SR, et al. A systematic review, meta-analysis and meta-regression of the effect of protein supplementation on resistance training-induced gains in muscle mass and strength in healthy adults. *Br J Sports Med.* 2018;52(6):376-384. PMID 28698222 (Correction: *Br J Sports Med.* 2020;54(19):e7. PMID 32943392. The correction adds a competing interest, not a change to the data: one author served on the advisory board of a sports supplement manufacturer while the paper was written. Corrected, not retracted; the breakpoint used here stands. Disclosed in the text.)
20. Lundh A, Lexchin J, Mintzes B, et al. Industry sponsorship and research outcome. *Cochrane Database Syst Rev.* 2017;2(2):MR000033. PMID 28207928

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Educational content only. Not medical advice. Not a diagnosis. Not a substitute for examination. The emergency signs at the top of this guide outrank everything else in it, the cardiac one especially. Talk with your clinician before starting new exercise or supplements, and respect the pressure ceilings above, which differ for manual tools and powered devices.